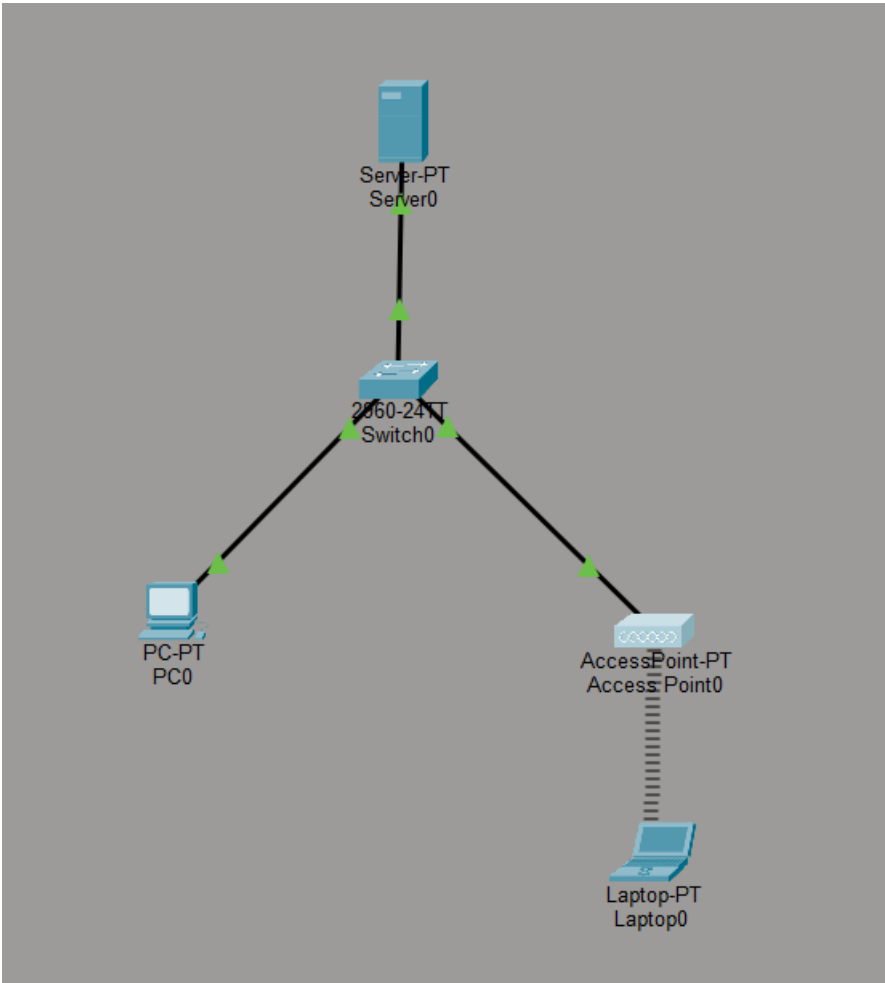


Q- Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation

To connect a laptop wirelessly to a WRT300N Wireless Router and understand the hardware requirements for wireless communication in Cisco Packet Tracer.

Network Topology



DHCP Server Configuration

Server IP Configuration

Parameter	Value
IP Address	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1

Server0

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

IP Configuration

DHCP

Static

IPv4 Address192.168.10.2

Subnet Mask255.255.255.0

Default Gateway192.168.10.1

DNS Server0.0.0.0

DHCP Pool:

Server0

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

DHCP

InterfaceFastEthernet0

Service

On

Off

Pool NameserverPool

Default Gateway192.168.10.1

DNS Server0.0.0.0

Start IP Address :

192

168

10

10

Subnet Mask:

255

255

255

0

Maximum Number of Users :245

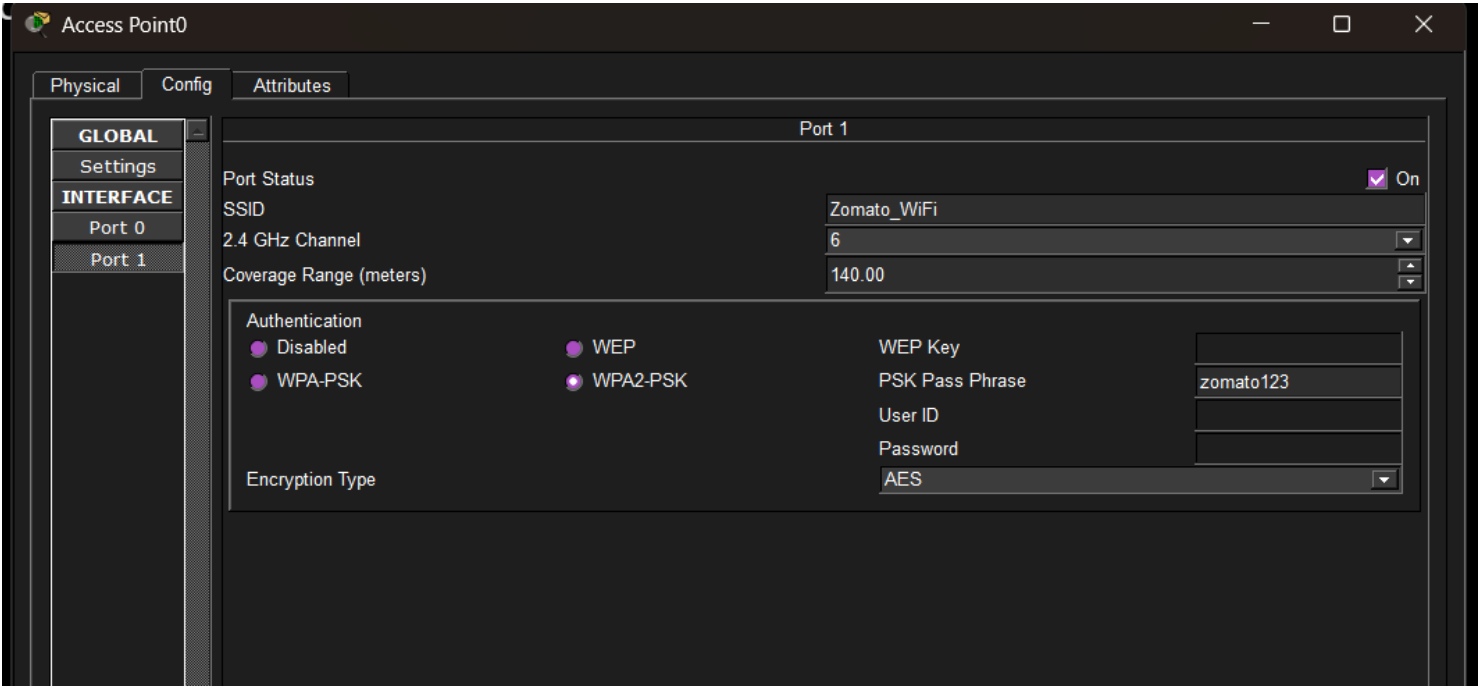
TFTP Server:0.0.0.0

WLC Address:0.0.0.0

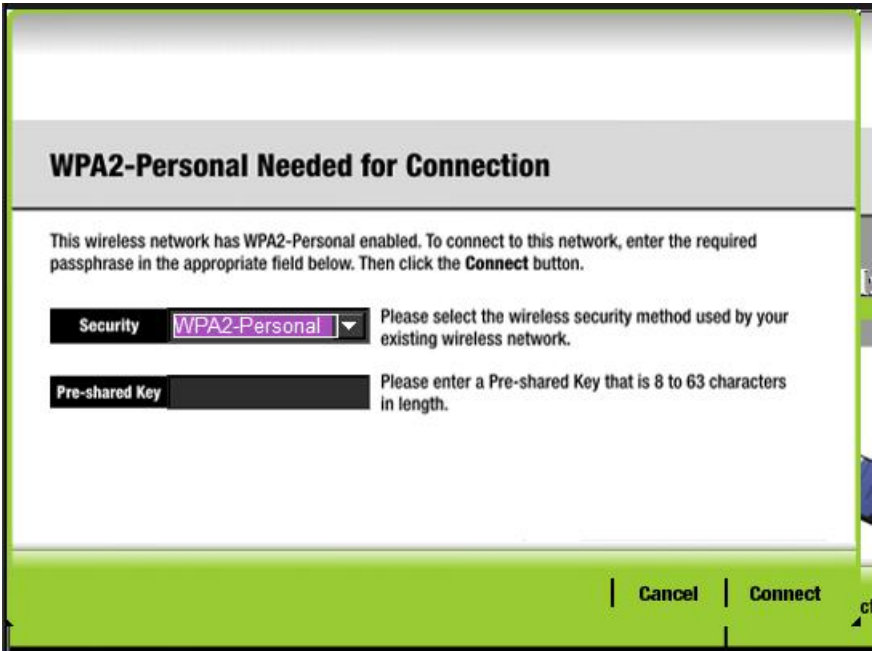
AddSaveRemove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.10.1	0.0.0.0	192.168.10.10	255.255.255.0	245	0.0.0.0	0.0.0.0

Access Point-Pt(Name and Password):



Laptop connect wireless



Laptop DHCP Ip address:

```
ipconfig

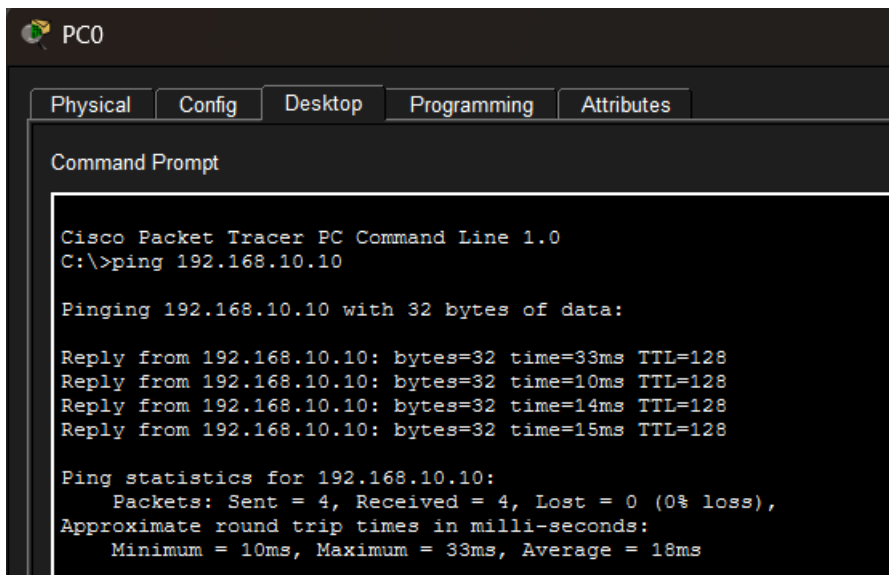
Bluetooth Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

Wireless0 Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::201:64FF:FE22:E60A
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.10.11
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                192.168.10.1
```

Check connectivity:



Why DHCP is Useful in a Restaurant WiFi Network

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses to devices when they connect to the network.

In a restaurant such as Zomato:

- Customers connect to WiFi without manually configuring IP addresses.
 - Each device receives a unique IP address automatically.
 - Network administration becomes easier.
 - IP address conflicts are prevented.
 - New customers can join the WiFi quickly and conveniently.
-

Advantages

- Automatic IP address allocation.
- Reduces manual configuration.
- Prevents duplicate IP addresses.
- Saves time for network administrators.
- Supports multiple users efficiently.